A Multi-Decadal Hourly Coincident Wind and Solar Power Production Dataset for the Contiguous United States

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ABSTRACT

As renewable energy continues its rapid expansion in the Unites States, multi-decadal hourly datasets of electricity production are needed to asses reliability and resource adequacy of power grids. Recent years have seen the release of grid-cell-level simulated meteorological variables, however these are not extended to the power domain, are not developed from a dynamically consistent numerical weather model, and only cover a historical baseline of less than a decade. To fill this gap, this work provides a dataset of 43 years of coincident plant-level wind and solar power production data. The dataset is designed to be aggregated to appropriate scales of interest for bulk system studies such as Balancing Authorities (BAs), states, and nodes of a production cost model. The dataset covers every plant in the contiguous U.S. that is reported in the U.S. Energy Information Administration (EIA) Form 860 as of 2020. When compared with the EIA-923 monthly generation, we find minimal bias (less than 5%). When compared with BA-reported hourly generation, we find low bias in solar (less than 7%), and slight underdispersion in wind. This coincident multi-decadal historical dataset provides a documented and evaluated multi-resource baseline for studies on reliability, resource adequacy, climate change impacts, and characterization of emergent climate threats on renewable resources.

Background & Summary

The electricity sector made up 25 percent of United States (U.S.) greenhouse gas (GHG) emissions in 2020[1]. In an effort to decarbonize the electricity sector, there is increased implementation of renewable energy sources, specifically wind and solar, which in 2020 made up 6 percent and 3 percent of global electricity generation, respectively[2]. This trend is expected to continue in the U.S., with falling costs[2], the production tax credit (PTC)[3] and investment tax credit (ITC)[4], and executive orders aiming at 100 percent carbon pollution-free electricity by 2030[5].

Wind and solar energy pose unique challenges for resource adequacy and reliability studies when compared to fossil-fuel based electricity generation, primarily due to their temporal variability[6]. In addition, wind and solar energy have high spatial variability, with more solar resources in the Southwestern U.S.[7] and more wind resources in the Great Plains and on the Pacific and Atlantic coasts[8]. These characteristics make power planning and grid operation significantly more complicated as renewable energy systems in the U.S. are non-dispatchable and are typically must-take resources at the scale of balancing authorities (BAs)[9], regions where generation and load must be balanced.

Compounding this, another challenge in power system planning is predicting the effects that climate change will have on solar and wind resources. Climate change is likely to impact the weather patterns that affect wind and solar resource potential. Gernaat et al. predict a complex pattern of changes to wind power in the 2070-2100 period across the U.S., with decreases in power production of as much as 20 percent in parts of the western U.S. due to decreasing wind speeds[10]. Pryor et al. predict a decline of up to 5 percent in annual wind energy density across the western U.S. by 2050, but an increase of 5-10 percent in the Great Plains region[11]. For solar power, Gernaat et al. predict a decrease of 0-5 percent in solar power in much of the western U.S. in 2070-2100 due to a decrease in solar irradiance[10], while Huber et al. show that annual global horizontal irradiance (GHI) will remain relatively constant in 2035-2039, while direct normal irradiance (DNI) will increase by up to 5 percent in the same time period in the U.S.[12].

Because the renewable fleet has evolved rapidly over the last two decades, reliability studies can only use the most recent years of observed wind and solar generation time series as a baseline. A small subset of years do not support robust benchmarks for understanding power grid operations under an evolving climate. The use of regional climate models (RCMs) to provide the meteorological input to develop power time series at contemporary power plants is the established state of the art approach[13–18] to support reliability studies under both historical and future conditions. In addition, to address research questions around climate change, it is critical that the climate projections be consistent with the historical benchmark simulations.

Previous studies and datasets have used machine learning models to predict wind and solar power generation based off historical generation data[19–21], as opposed to using climate models to predict power generation. However, due to the climate change projections described above, machine learning models will be unable to capture the changes in climate that will affect renewable energy production. Due to these projections of nonstationarity in weather patterns, there is increased urgency to develop models to accurately predict renewable energy generation from climate models.

Established approaches exist to translate high resolution climate model simulations into power time series. Studies that have used climate models to calculate wind and solar power generation typically have used one or more baseline technologies for calculation of wind and solar power[15,22,23], which fails to capture the differences in technologies that impact power generation (i.e., wind turbine power curve and hub height, solar PV tilt angle, and module type). Wang et al. used the customized wind turbine power curves for each plant to calculate power generation, but only did so for five wind farms in California[24]. Campbell leveraged nine representative default power curves[25] to produce wind power generation for all current wind plants in the U.S., but only for the year 2018[26]. Millstein et al. produced wind power generation profiles for the same set of current U.S. wind plants with publicly accessible power curves (PLUSWIND), expanding the available historical baseline to four years[27]. The use of customized power plant configurations is a necessary innovation in the development of historical benchmark wind and solar power time series which help to more realistically evaluate the impacts of climate change and technology innovation.

Acceptance and trust in synthetic power production datasets derived from climate models hinges on comprehensive validation. Moncheur de Rieudotte et al. supported this with the recent release of the wind energy and power validation toolset, WE-Validate, demonstrating its ease of use on their synthetic wind power datasets for 2018[26,28]. To illustrate the limitations of current reanalysis products, Davidson et al. validated synthetic wind power produced from three sources, showing that diurnal hourly power plant-level profile accuracy and correlation declines during evening hours[29]. While these validation studies are crucial to improving the acceptance of synthetic generation datasets, they were not demonstrated on coincident wind and solar generation, nor at the BA scale, which is critical for assessment of system-wide generation and load balancing needs. The contribution of this work is a thorough evaluation across scales and features that make subsequent datasets with a future infrastructure or weather realistic and trustful with transparency on their accuracy.

We present a new wind and solar dataset that uses regional atmospheric climate model simulations of a historical baseline and 2020 wind and solar power plant configurations across the entire contiguous U.S. This dataset is developed at the plant scale but designed to be aggregated to the BA scale (i.e., regional power delivery) or to other scales such as nodes of production cost models, which allow the dataset to be used to perform reliability assessments and evaluations of technology innovation at system scales. The choice of aggregation to the BA level was made to reflect the scale in the U.S. where wind and solar are balanced with load before importing external resources, which is necessary during extended periods of resource droughts[30]. This dataset serves as a multi-decadal record for current wind and solar infrastructure and can be used to understand sensitivity to historical interannual variability, seasonality and recent extreme events. In addition it provides a baseline for future projections of renewable resources. These data were produced under the Pacific Northwest National Laboratory's (PNNL) Grid Operations, Decarbonization, Environmental, and Energy Equity Platform (GODEEEP). The wind and solar datasets described in this paper will be referred to as GODEEEP-wind and GODEEEP-solar, respectively. We present below the approach and the evaluation of the datasets across multiple scales to support their use in power system studies.

Methods

The general process for creating generation time series is (1) develop power plant specifications from the Form EIA-860 database, to be used as input parameters to the power generation model, (2) preprocess the meteorological data to be suitable for input into the power generation model, and (3) run the power generation model and postprocess the output. This general process is identical for both wind and solar. This section describes each of these steps in detail.

Power Plant Specifications

The Form EIA-860[31] is published annually by the Unites States Energy Information Administration. The database contains generator level information for power plants with a nameplate capacity of 1 MW or greater (Figure 1). This database is the primary source of information we use to develop input parameters to the power generation model. While the Form EIA-860 database is a valuable source of information, it has significant limitations, specifically for wind power. Namely, EIA only tracks the 'Predominate Turbine Model Number' and the overall number of turbines, rather than all models and the corresponding number of turbines for each model. This means that for generators that have more than one unique turbine model present, the EIA data does not accurately represent the true fleet of wind turbines. To ensure the most accurate power generation calculations, the configuration files were created using the 'Predominate Turbine Model Number' and adjusting the number of turbines to enforce the relationship

$$Capacity_{Plant} = N_{Turbines} * P_{Turbine,Rated}. \quad (1)$$

The specifications for the given turbine model are found at wind-turbine-models.com[32] and in the System Advisory Model (SAM)[33], and the turbine coordinates are found in the U.S. Wind Turbine Database[34]. In addition, some assumptions are necessary to build a complete set of input parameters for the power generation model, which will be discussed below.

Power generation models require certain input parameters for each power plant; we refer to these collectively as the plant configuration. The EIA-860 configuration parameters are provided at the generator level (e.g., number of turbines, primary turbine make/model), where a plant can be a collection of one to many generators. We aggregated generator level information to the plant level. Joining the plant data to the generator data also allows the user to filter to a specific North American Electric Reliability Corporation (NERC) region. We collected all of the plant configurations into a csv file (one per technology type) where each row corresponds to a power plant and the column headers refer to model input parameters or other plant metadata. The ensuing steps are unique for each technology, and are described in detail below.

Solar

The required inputs for the solar plant configurations are shown in Table 1. The system capacity corresponds to the "Nameplate Capacity (MW)" column in Form EIA-860, converted from megawatts to kilowatts. The following rules were used to determine the module type, array type, azimuth, and tilt angle from the Form EIA-860. Note that names listed below in quotes such as "Fixed Tilt?" refer to columns in the Form EIA-860 solar database.

1. Array Type
   (a) Plants listed with a "Y" in the "Fixed Tilt?" column are classified as fixed open rack arrays.
   (b) Plants listed with a "Y" in the "Single-Axis Tracking?" are classified as 1-axis tracking arrays.
   (c) Plants listed with a "Y" in the "Dual-Axis Tracking?" are classified as 2-axis tracking arrays.
   (d) Any plants without a listed array type are classified as fixed open rack arrays.
2. Module Type
   (a) Plants listed with a "Y" in the "Thin-Film (CdTe)?", "Thin-Film (A-Si)?", "Thin-Film (CIGS)?", or "Thin-Film (Other)?" column are classified as thin film modules.
   (b) Any other plants are classified as standard modules.
3. Azimuth Angle
   (a) For any plant with a positive "Azimuth Angle", the value was used as is.
   (b) For any plant with a negative "Azimuth Angle", the value was assumed to be an entry error, in which case the absolute value was used.
   (c) For any plant without a listed "Azimuth Angle", the values was assumed to be 180.
4. Tilt Angle
   (a) For any plant with a positive "Tilt Angle", the value was used as is.
   (b) For any plant with a negative "Tilt Angle", the value was assumed to be an entry error, in which case the absolute value was used.
   (c) For any plant without a listed "Tilt Angle", the values was assumed to be the latitude of the plant.

The losses are set to 14%, the default value used in PVWatts[35]. The Form EIA-860 database contains many plants with the same plant code that are split up across multiple rows. Where possible the entries with the same plant code were combined if all the plant configuration parameters were identical. When combining was not possible each entry in the database with the same plant code was modeled as a separate plant. Finally, these specifications are output in a .csv file with one row entry for each plant.

Wind

The wind configuration file generation process begins by importing the nameplate capacity and the turbine hub height from the Form EIA-860 wind database. The remaining required specifications aren't contained in the Form EIA-860 database. We used a combination of turbine model specifications found in SAM[33] and publicly available databases[32] to determine rotor diameter and the power curves. The U.S. Wind Turbine Database (USWTDB)[34] contains wind turbine coordinate sets for each plant. For wake calculations, we used the Simple Wake Model, originally developed by the University of Wisconsin's Solar Energy Research Center[36] and adapted for use by NREL. This model estimates the reduction in wind speed at a downwind turbine due to the wake of an upwind turbine using the wind speed deficit factor[37]. The wind resource shear and wind resource turbulence coefficient are set to the default values of 0.14 and 0.1, respectively. Wind model input variables are described in Table 2.

Wind turbine power curves (wind speeds and corresponding power outputs) are available for many models. For those that do not have power curves available, an approximation was developed using the cut-in, rated, and cut-out wind speeds along with the rated power of the wind turbine. A power curve was developed using these parameters to fit a logistic growth equation[38].

The coordinates of the individual wind turbines at a given plant are available for some plants in the USWTDB. Because wind power models calculate the plant power output by summing up the power output of the individual turbines provided in the coordinate sets, it is important that the number of coordinates provided multiplied by the rated power of each turbine is equal to the nameplate capacity of the plant (a threshold of 5 percent was used), as defined by Equation 1. Many of the plants do not have the correct number of coordinates – for those with an incorrect number of coordinates, a rectangular grid layout was assumed with sizing based on the rotor diameter. The number of coordinates used is computed by dividing the nameplate capacity of the plant by the rated power of each turbine. Each row and column of the grid is spaced apart by eight rotor diameters, and every other row offset by four rotor diameters, as done in the default wind farm layout in SAM[33].

Meteorological data preprocessing

The meteorological input data used is from the Thermodynamic Global Warming (TGW) simulations[39,40]. TGW is a 1/8° dynamically downscaled product available for the conterminous Unites States that is intended to replicate historical weather patterns (1980-2019) into future periods. While it is not precisely a reanalysis product over the historical period (updated to 1980-2022 in this work), it is expected to provide reasonably accurate results at aggregated scales such as BAs or states, which is the intended purpose of this dataset. Surface data that is appropriate for input to a solar generation model is available at 1-hour resolution while wind model inputs are available at 3-hour resolution.

Wind power models need weather data inputs at specific hub heights though TGW uses a vertical coordinate represented by pressure levels, which is typical for weather and climate models. In order to derive meteorological data at specific heights above ground level, the pressure levels must be interpolated. This vertical interpolation was done using the wrf-python package[41]. We produced vertically interpolated wind input variables at the hub heights for each turbine. In addition the 3-hourly upper level atmospheric data is interpolated to hourly.

Solar power models typically need three components of solar irradiance: global horizontal irradiance (GHI, aka shortwave downward radiation), diffuse normal irradiance (DNI) and diffuse horizontal irradiance (DHI). These variables are related by the equation $\mathrm{GHI} = \mathrm{DNI}\cos(\theta) + \mathrm{DHI}$, where θ is the solar zenith angle at a particular location. TGW only produces GHI so we compute DNI using the Direct Insolation Simulation Code (DISC)[42,43]. DISC is a physically based statistical model for computing DNI from GHI and is available as part of the NREL farms package[44].

Wind and Solar generation model

To produce power from meteorological inputs we used the NREL renewable energy potential (reV) model[45,46]. The reV model uses code from the PySAM package[47] which in turn uses the same core code as the System Advisor Model (SAM). The individual models available in these packages are all NREL developed and widely used in the industry. For wind we use the windpower model[37] and for solar we use the PVWatts V7[35] model. The meteorological inputs required for the wind model are temperature (C), pressure (Pa), wind direction (degrees), and wind speed (m/s) at the hub height of the turbine. The meteorological inputs to the solar model are temperature (C), solar irradiance (at least 2 of GHI, DNI, or DHI) (W/m^2), pressure (hPa), and wind speed (m/s) at the surface. The reV configuration input variables and sources are shown in Table 1 for solar and Table 2 for wind. The outputs of the models are hourly power generation expressed as capacity factor (generation divided by plant capacity). The reV model produces 8,760 points for each year, where the last day in the year is truncated for leap years.

Data Records

The datasets consist of 43 years (1980-2022) of simulated historical plant level hourly capacity factors with 2020 plant configurations. For purpose of evaluation, we also developed 13 years of balancing authority level aggregated hourly power production and 13 years of monthly plant level inventories for each balancing authority, both associated with the evolving

2007-2020 wind and solar infrastructure. This 13 year period corresponds to the availability of the observed wind and solar datasets.

The historical hourly capacity factor profiles for wind and solar are provided on Zenodo[48] in csv format with one file per year per technology type, e.g. `[technology]_gen_cf_[yyyy].csv` (https://zenodo.org/record/8240163). The first column in each of these files is the datetime in UTC. Subsequent columns are named based on the Form EIA-860 plant codes. In cases where multiple rows in the EIA database could not be combined, the EIA-860 plant code is appended with an index, producing unique plant codes. Each row in the file corresponds to one hour. Data values are expressed as capacity factor, where hourly generation is divided by installed plant capacity.

The plant configuration files required to convert meteorological inputs to power with reV are provided in the same repository as the capacity factor profiles in csv format, `eia_wind_configs.csv` and `eia_solar_configs.csv`[48]. Each line in the file represents one modeled plant. Cases where a single plant can not be modeled at once are plants with multiple panel types (solar plant) or multiple turbine types (wind plant).

The validation datasets, which include the 13 year historical balancing authority level aggregated power and plant inventories, are provided on Zenodo[49] in csv format with one file for each technology type at the balancing authority level and at the plant level (https://zenodo.org/record/8325956). The plant level column names are in the format `PlantID_GeneratorID` following the naming technique in EIA-860. The monthly balancing authority installed inventory is found in the file `all_years_860m.csv`, with columns containing the plant id, generator id, plant name, nameplate capacity in that month, balancing authority in that month, resource technology (wind or solar), month, year, `plant_code_unique` to match with the capacity factor files, `scale_nameplate` to indicate whether the plant needed to be split into two configuration rows for use in reV, and `plant_gen_id` to align with the column headers in the plant level files.

Technical Validation

Meteorology

While full validation of the TGW data is out of the scope of this paper, we conducted validation of a few variables that are important for wind and solar power production. Full TGW validation results are provided in[39]. For solar, we compared GHI and surface temperature against the NREL National Solar Radiation Database (NSRDB)[50]. While surface temperature showed little bias, GHI showed a 10-15% high bias compared to NSRDB. To account for the GHI bias, we bias corrected the resulting solar power data using a quantile mapping approach[51] using the full NSRDB data from 1998 to 2000 which we converted to solar power with the same plant configurations. Both bias corrected and non bias corrected power profiles are provided in the final dataset. For wind, observational datasets are only available at a handful of locations and with inconsistent historical coverage. ERA5, from which the TGW data is derived, shows negative bias in wind speeds in the available central U.S. observation locations, however the lack of gridded observations (as available for GHI with the NSRDB) impedes consistent bias correction[52]. To address the continent-wide evaluation of wind variables, we compared the U and V component wind speeds, pressure, and temperature in the TGW data to those in the gridded NREL WIND Toolkit (WTK). All variables showed less than 5% average bias, so no bias correction was conducted.

Power

The validation of the power time series is conducted against plant-level monthly energy (EIA-923) and BA-level self-reported hourly production (referred to as “scada”). To facilitate this comparison, the normalized (capacity factor) power time series were aggregated to the BA level and converted to actual production by applying the monthly installed capacity according to the EIA-860 monthly generator inventory. This approach captures the infrastructure evolution in each BA at a monthly scale.

The synthetic hourly wind and solar power time series are compared against two publicly available datasets: EIA-923 monthly plant-level energy (MWh)[53] and BA-level hourly scada (MW). The EIA-923 provides accurate plant-level reporting against which the plant-to-BA-level monthly aggregations can be compared, while the BA-level scada provides a comparison of hourly production. Both power validation sources have available data beginning in 2007, with some records starting in later years consistent with infrastructure build out. Scada data is available for different baseline periods for each comparison BA ranging from 2007 to 2020. The data availability for each BA largely tracks with resource adoption. The evaluation of the wind and solar datasets against observational data can be performed over only recent years given the fast expansion over the last two decades, however this short period provides large caveats: it does not address the accuracy of synthetic power datasets in hot or cold years and fundamentally assumes that weather drives inter-annual variability over the fast evolution of the fleet. For these reasons, this work targets applications for regional power grid studies which must assess the need to support system generation balanced with load.

We validate the hourly wind and solar production for (1) annual BA-level infrastructure growth, (2) seasonal variability within NERC regions, and (3) seasonal shape within NERC regions and diurnal shape within BA. The BAs we examine are BPAT (Bonneville Power Authority)[54], CISO (California ISO)[55], ERCOT (Electric Reliability Council of Texas)[56], ISONE

(ISO of New England)[57], MISO (Midwest ISO)[58], and SWPP (South West Power Pool)[59]. Figure 1 shows the BA control areas from Homeland Infrastructure Foundation-Level Data (HIFLD) which approximate BA areas. The NERC interconnection regions are the Eastern Interconnect (EIC), the Electric Reliability Council of Texas (ERCOT), and the Western Interconnect (WECC).

Infrastructure Build-Out

Given known BA wind and solar capacity in a particular year, a first evaluation is to accurately reproduce the compound infrastructure buildout and inter-annual variability over time when compared EIA-923 and scada. Figure 2 shows annual solar (left column) and wind (right column) for five BAs. Despite some bias when compared to EIA-923 and scada data, the overall trend in infrastructure development is captured well in every BA. For wind, some notable high bias is present in ERCOT while CAISO shows some low bias. The meteorological validation we conducted revealed a small positive bias in wind speed with respect to the WIND Toolkit, suggesting that localized correction of the bias in ERCOT could contribute to reducing the bias in the wind power. The WTK provided a comparison of average daily production with ERCOT and with MISO, showing that it too is biased high compared with the reported BA-level measurements[60]. The PLUSWIND dataset also shows a high bias of wind power production in ERCOT, in agreement with our synthetic wind power production[27]. Given the limited availability of wind speed observations, we provide the wind power profiles without bias corrections. For solar, CISO and ERCOT show a notable low bias.

Seasonal Variability

To examine the seasonal variability we aggregated all the generation across the three the NERC regions, the Eastern Interconnect (EIC), the Electric Reliability Council of Texas (ERCOT), and the Western Interconnect (WECC). Figure 3 shows the seasonal variability of GODEEEP-solar (left column) and GODEEEP-wind (right column) vs EIA-923, where the boxes represent the variability across all available years. The seasonal shape is generally well represented with some mismatch occurring in the variability. This indicates that the TGW data is accurately capturing the seasonal climatological patterns and those are being translated appropriately to the generation data. The mismatch in variability could be due to inaccuracies in the EIA-923 data or misreporting in the plant characteristics in EIA-860, and assumptions made during development of the plant configurations. Bias tends to be low, 5% or less.

Diurnal Shape

The final aspect of the wind and solar data we examine is the diurnal shape, which is particularly important to capture for grid reliability. Figure 4 shows ERCOT solar (left column) and BPAT wind (right column) for the entire year (top row), the spring (MAM, middle row), and the fall (SON, bottom row) seasons. Overall the diurnal patterns are well represented in each season and across all the entire year. For solar we generally see a low bias (up to 7%) compared to the SCADA data in ERCOT – this is likely due to a combination of errors in the scada data and inaccuracies in the plant configuration assumptions, for example, we use a flat 14% loss assumption. For wind, the results are generally favorable with a slight underdispersion apparent in most hours.

Usage Notes

The wind and solar data files provided in[48] are available in csv format and therefore should be readable by all commonly used software platforms. If the users wish to work with generation data instead of capacity factor, one way to do the conversion for all plants at once is to transform the data into long format with three columns, `datetime`, `plant_code`, and `cf` (the capacity factor), then join the appropriate wind/solar configuration file by `plant_code` and finally multiply the `cf` column by the `system_capacity` column to get the power generation in kW. Please also note that the balancing authorities that are associated with each plant are those reported as of 2020. Plants may change balancing authorities so this classification is only valid for this dataset.

Code availability

The hourly wind and solar capacity factor datasets developed for the 1980-2022 period and the EIA 2020 plant configuration (`eia_solar_configs.csv` and `eia_wind_configs.csv`) are available at https://zenodo.org/records/8393319. The code to reproduce the capacity factor power profiles is available at https://github.com/GODEEEP/tgw-gen. The validation datasets, including the hourly wind and solar power generation datasets developed for the 2007-2020 period representative of monthly infrastructure build-out at the BA scale and the monthly mapping of power plant to BA, are available at https://zenodo.org/records/8325956. The code to produce the hourly BA-level power containing monthly infrastructure growth, for use in evaluating the capacity factor datasets, is available at https://github.com/GODEEEP/renewable-power-evaluation.

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Author contributions statement

A.C. conceived of the workflow. C.B. wrote the code to develop the wind and solar dataset. S.U. wrote the code to develop the plant configurations. C.B., S.U. and A.C. wrote the first draft of the manuscript. A.C. conducted the validation. N.V. provided input and guidance on the workflow, manuscript framing, and validation as well as edited the manuscript.

Competing interests

The authors declare no competing interests.

Figures & Tables

Input Variable	Source
Plant Capacity	Form EIA-860 'Nameplate Capacity (MW)'
Array Type	Form EIA-860 (Multiple Fields)
Azimuth Angle	Form EIA-860 "Azimuth Angle"
Module Type	Form EIA-860 (Multiple Fields)
Tilt Angle	Form EIA-860 "Tilt Angle"
Losses	14%

Table 1. Input variables and their respective sources for solar configuration files using PVWatts[35].

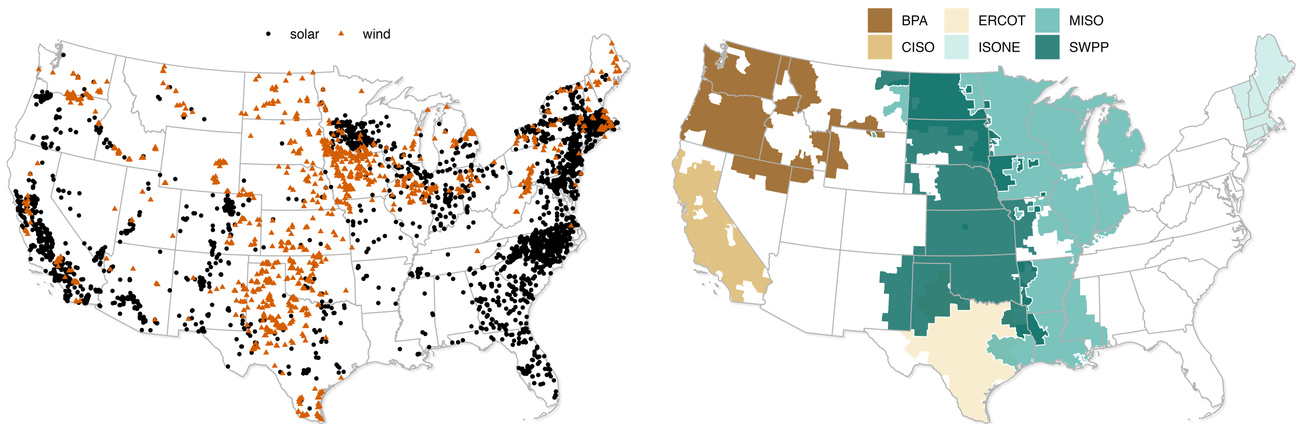

Figure 1. Wind and Solar plant locations provided by this dataset (left) and approximate Balancing Authority (BA) areas used for validation in this study (right). In practice plants are assigned to BAs based on their entry in the EIA-860 database not based on these geographic areas.

Input Variable	Source
Plant Capacity	Form EIA-860 'Nameplate Capacity (MW)'
Wake Model	Simple
Turbine Coordinates	U.S. Wind Turbine Database
Shear Coefficient	0.14
Turbulence Coefficient	0.1
Hub Height	Form EIA-860 'Turbine Hub Height (Feet)'
Power Curve	Published power curve or approximated
Rotor Diameter	wind-turbine-models.com[32] and SAM[33]
Losses	14%

Table 2. Input variables and their respective sources for wind configuration files using the PySAM Windpower module[47].

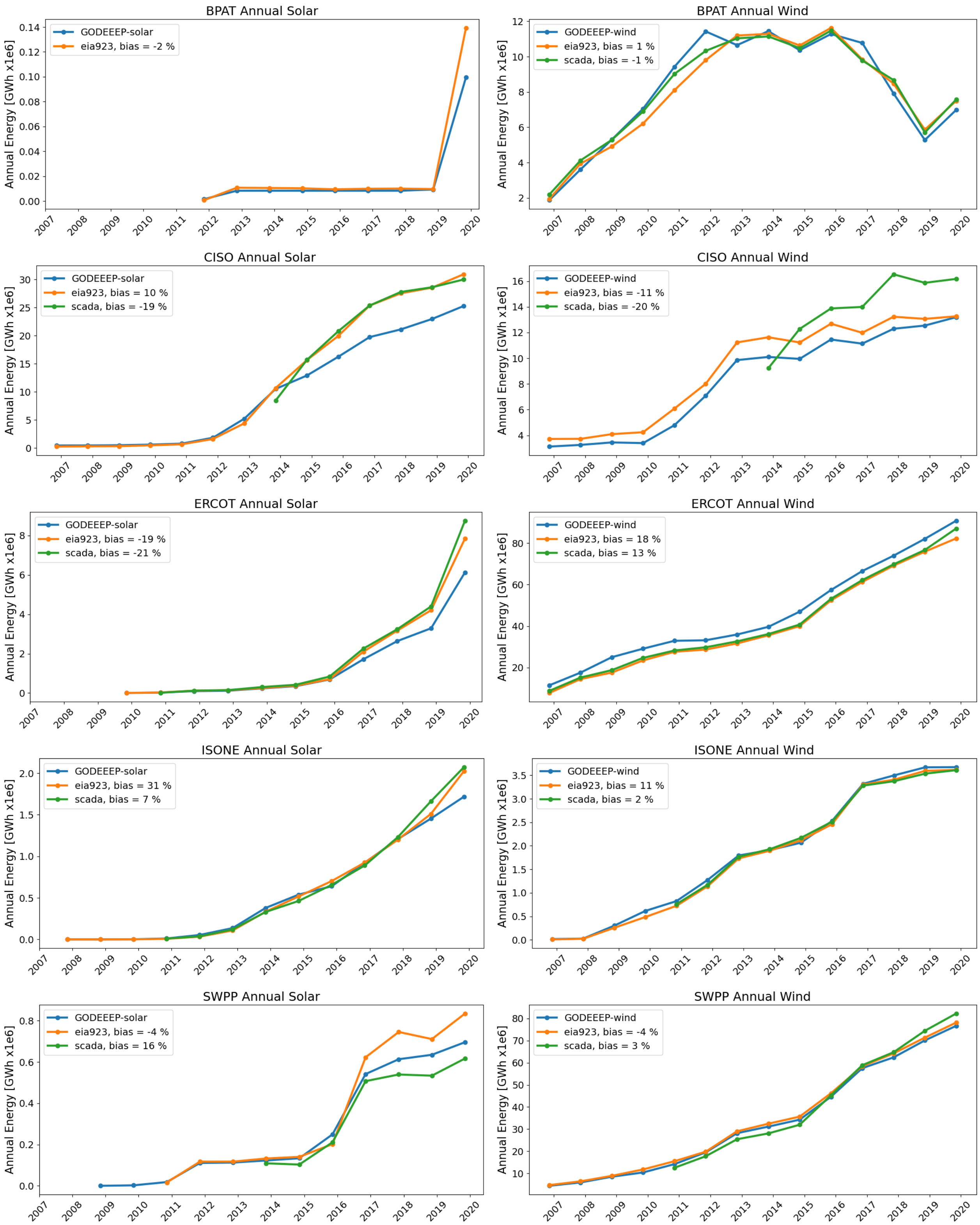

Figure 2. The annual wind energy delivered by BA shows the annual change in infrastructure. The EIA-923 and GODEEEP datasets are aggregated to the BA according to the EIA-860 operating reports while SCADA come pre-aggregated by the BA. The percent difference with respect to GODEEEP is averaged over the number of years that overlap with the comparison datasets EIA-923 and BA self-reported SCADA.

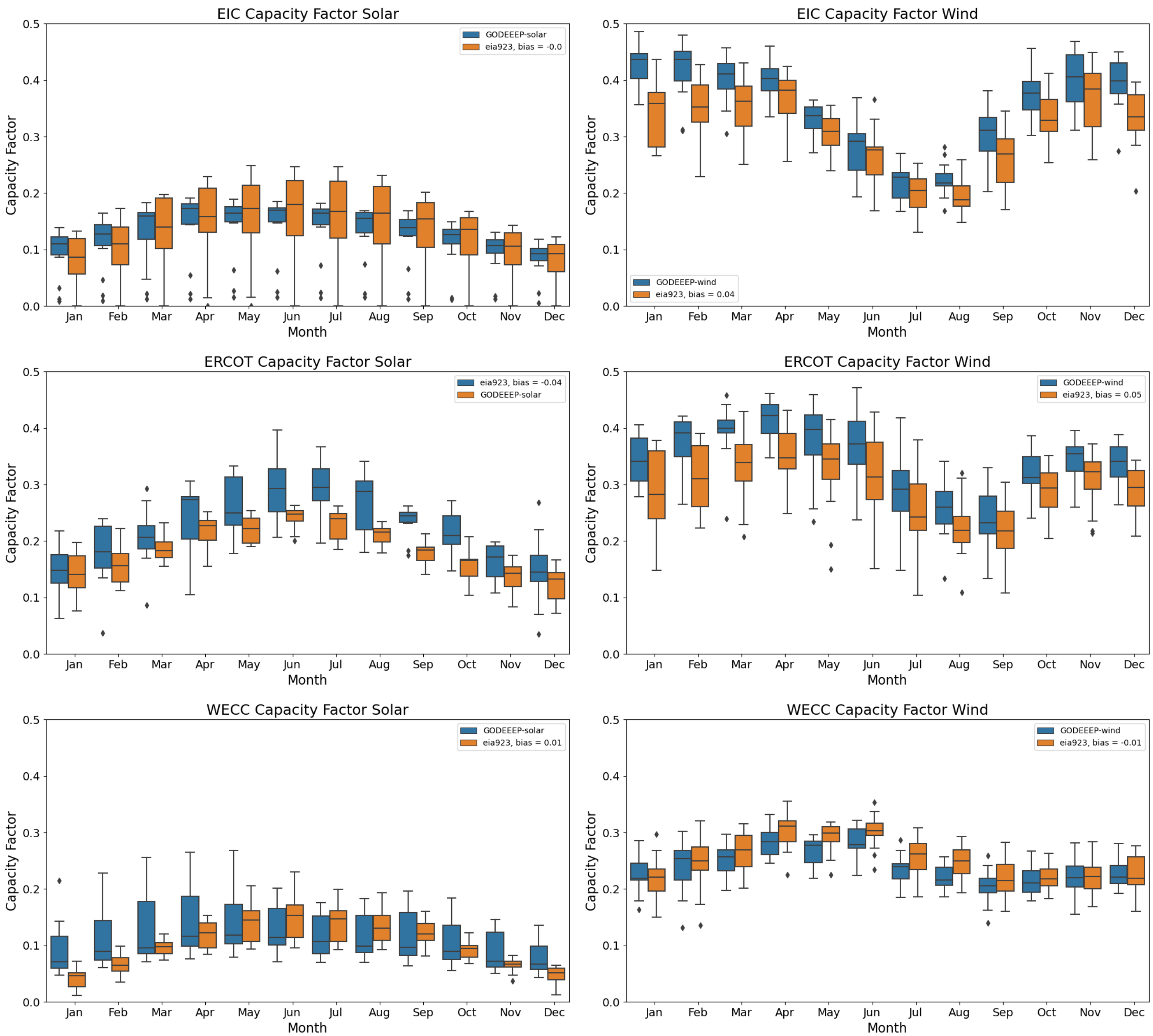

Figure 3. Monthly capacity factor in each NERC region for wind (left) and solar (right). Monthly energy production has been normalized to the total installed capacity in each month and year for each BA in the NERC region according to EIA860 reports. Bias with respect to EIA923 is averaged across all months.

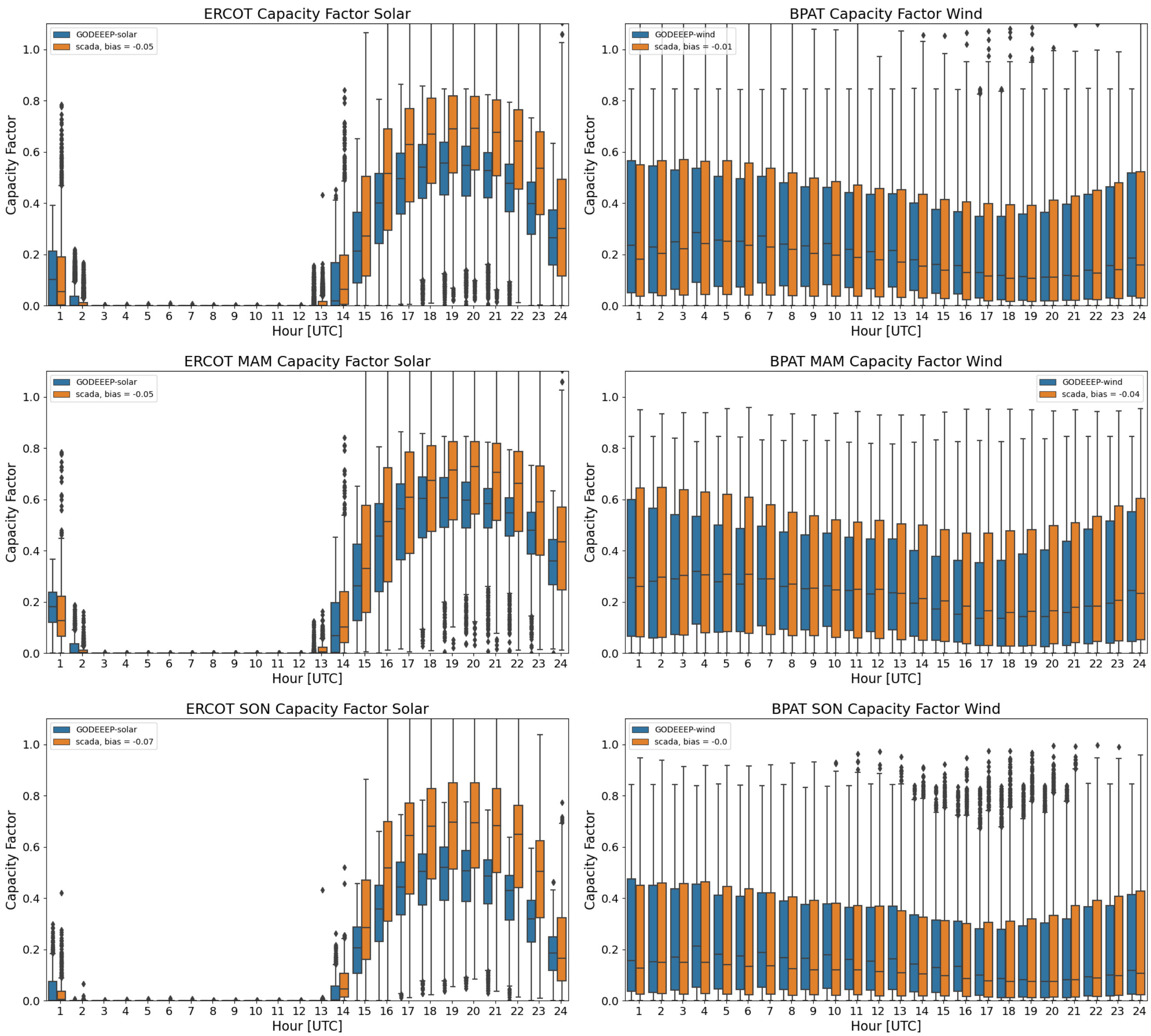

Figure 4. Diurnal capacity factor for ERCOT Solar (left) and BPAT Wind (right) relative to BA-reported SCADA. Capacity factors are calculated by normalizing hourly production with respect to EIA860 reports of BA operational power plants. The annual diurnal shapes (top row) represent the capacity factor in each hour over the full historical baseline of SCADA data. The middle and bottom rows segment the annual diurnal shapes by season, with spring March April May (MAM, middle row) and fall September October November (SON, bottom row). Capacity factor values greater than one for SCADA reflect inaccuracies in reported operational BA-level infrastructure in EIA860.